lab exp 3 | Home | Untitled9

localhost:8888/notebooks/Untitled9.ipynb

jupyter Untitled9 Last Checkpoint: 1 minute ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel)

[1]:
```python
# ==========================================
# FAST Handwritten Digit Recognition
# Using Linear SVM and KNN
# ==========================================

# Import Required Libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.svm import LinearSVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load MNIST Dataset
print("Loading MNIST dataset...")
mnist = fetch_openml('mnist_784', version=1)

X = mnist.data
y = mnist.target.astype(int)

# Reduce dataset size for faster execution
X = X[:10000]
y = y[:10000]

print("Reduced Dataset Shape:", X.shape)

# Data Preprocessing
# Normalize pixel values
X = X / 255.0

# Train-Test Split
```

29°C Sunny | Search | ENG IN | 10:21 AM 26-03-2026

lab exp 3 | Home | Untitled9

localhost:8888/notebooks/Untitled9.ipynb

jupyter Untitled9 Last Checkpoint: 1 minute ago

File Edit View Run Kernel Settings Help

Code

```
print("Reduced Dataset Shape: ", X.shape)

# Data Preprocessing
# Normalize pixel values
X = X / 255.0

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ========================================
# Linear Support Vector Machine
# ========================================
print("\nTraining Linear SVM...")
svm = LinearSVC(max_iter=5000)
svm.fit(X_train, y_train)

y_pred_svm = svm.predict(X_test)

print("\n===== Linear SVM RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_svm))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_svm))
print("Classification Report:\n", classification_report(y_test, y_pred_svm))

# ========================================
# K-Nearest Neighbors
# ========================================
print("\nTraining KNN...")
knn = KNeighborsClassifier(n_neighbors=5, n_jobs=-1)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

print("\n===== KNN RESULTS =====")
```

lab exp 3 | Home | Untitled9

localhost:8888/notebooks/Untitled9.ipynb

jupyter Untitled9 Last Checkpoint: 1 minute ago

File Edit View Run Kernel Settings Help

Code

```
print("\n===== Linear SVM RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_svm))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_svm))
print("Classification Report:\n", classification_report(y_test, y_pred_svm))

# ========================================
# K-Nearest Neighbors
# ========================================
print("\nTraining KNN...")
knn = KNeighborsClassifier(n_neighbors=5, n_jobs=-1)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

print("\n===== KNN RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_knn))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred_knn))
print("Classification Report:\n", classification_report(y_test, y_pred_knn))

# ========================================
# Visualize Sample Predictions
# ========================================
plt.figure(figsize=(10, 5))

for i in range(10):
    plt.subplot(2, 5, i + 1)
    plt.imshow(X_test.iloc[i].values.reshape(28, 28), cmap='gray')
    plt.title("Pred: " + str(y_pred_knn[i]))
    plt.axis('off')

plt.suptitle("Sample KNN Predictions")
plt.show()
```

```
Loading MNIST dataset...
```

JupyterLab Python 3 (ipykernel)

29°C Sunny | Search | ENG IN | 10:22 AM 26-03-2026

lab exp 3 | Home | Untitled9

localhost:8888/notebooks/Untitled9.ipynb

jupyter Untitled9 Last Checkpoint: 1 minute ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel)

```python
    plt.axis('off')

plt.suptitle("Sample KNN Predictions")
plt.show()
```

```
Loading MNIST dataset...
Reduced Dataset Shape: (10000, 784)

Training Linear SVM...

===== Linear SVM RESULTS =====
Accuracy: 0.9
Confusion Matrix:
 [[202   0   1   0   0   1   1   0   2   0]
 [  0 210   2   0   0   1   0   1   2   0]
 [  4   3 169   8   1   3   4   1   9   2]
 [  0   1   5 165   0   9   2   3   5   2]
 [  0   0   3   0 191   2   4   1   1   9]
 [  4   3   3  11   1 148   2   0   3   1]
 [  1   0   2   0   0   5 210   0   1   1]
 [  0   1   2   2   3   1   0 202   0   5]
 [  0   2   1  11   2   3   0   3 139   5]
 [  2   0   2   5   2   1   1  13   2 164]]
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.98      0.96       207
           1       0.95      0.97      0.96       216
           2       0.89      0.83      0.86       204
           3       0.82      0.86      0.84       192
           4       0.95      0.91      0.93       211
           5       0.85      0.84      0.85       176
           6       0.94      0.95      0.95       220
           7       0.90      0.94      0.92       216
           8       0.85      0.84      0.84       166
           9       0.87      0.85      0.86       192
```

29°C Sunny | Search | ENG IN | 10:22 AM 26-03-2026

lab exp 3 | Home | Untitled9

localhost:8888/notebooks/Untitled9.ipynb

jupyter Untitled9 Last Checkpoint: 1 minute ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel)

```
    accuracy                           0.90      2000
   macro avg       0.90      0.90      0.90      2000
weighted avg       0.90      0.90      0.90      2000


Training KNN...

===== KNN RESULTS =====
Accuracy: 0.9475
Confusion Matrix:
 [[206   1   0   0   0   0   0   0   0   0]
 [  0 216   0   0   0   0   0   0   0   0]
 [  2   5 192   0   0   0   0   4   0   1]
 [  1   2   1 178   0   5   0   3   1   1]
 [  1   3   0   0 196   0   2   0   0   9]
 [  2   2   0   4   1 164   1   1   0   1]
 [  3   0   0   0   0   2 215   0   0   0]
 [  0   5   0   0   2   0   0 207   0   2]
 [  0   3   1   3   1   7   3   2 142   4]
 [  3   1   0   3   0   0   0   6   0 179]]
Classification Report:
              precision    recall  f1-score   support

           0       0.94      1.00      0.97       207
           1       0.91      1.00      0.95       216
           2       0.99      0.94      0.96       204
           3       0.95      0.93      0.94       192
           4       0.98      0.93      0.95       211
           5       0.92      0.93      0.93       176
           6       0.97      0.98      0.98       220
           7       0.93      0.96      0.94       216
           8       0.99      0.86      0.92       166
           9       0.91      0.93      0.92       192

    accuracy                           0.95      2000
   macro avg       0.95      0.94      0.95      2000
```

29°C Sunny | Search | ENG IN | 10:22 AM 26-03-2026

lab exp 3 | Home | Untitled9

localhost:8888/notebooks/Untitled9.ipynb

jupyter Untitled9 Last Checkpoint: 2 minutes ago

File Edit View Run Kernel Settings Help

Trusted

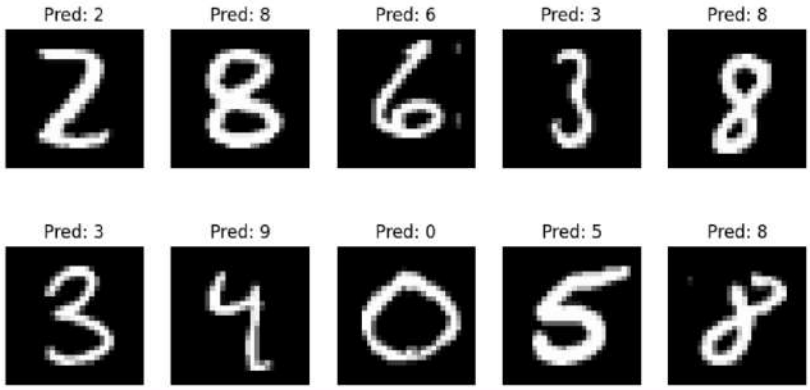
Code

JupyterLab Python 3 (ipykernel)

```
           6       0.97      0.98      0.98       220
           7       0.93      0.96      0.94       216
           8       0.99      0.86      0.92       166
           9       0.91      0.93      0.92       192

    accuracy                           0.95      2000
   macro avg       0.95      0.94      0.95      2000
weighted avg       0.95      0.95      0.95      2000
```

Sample KNN Predictions



29°C Sunny | Search | ENG IN | 10:22 AM 26-03-2026